

2. 8-PUZZLE PROBLEM

Aim

To write a Python program to solve the 8-Puzzle problem.

Objective

To arrange the tiles in the correct order by moving the blank space.

Algorithm

1. Define the initial state and goal state.
2. Find the blank space.
3. Move the blank tile to a valid position.
4. Repeat until the goal state is reached.
5. Display the solution.

Python code:

```
from collections import deque
```

```
start = tuple(map(int, input("Enter start state (9 numbers separated by space):").split()))
```

```
goal = tuple(map(int, input("Enter goal state (9 numbers separated by space):").split()))
```

```
moves = {  
    0:[1,3], 1:[0,2,4], 2:[1,5],  
    3:[0,4,6], 4:[1,3,5,7], 5:[2,4,8],  
    6:[3,7], 7:[4,6,8], 8:[5,7]  
}
```

```
queue = deque([(start, [])])
```

```
visited = set()
```

```
while queue:
```

```
state, path = queue.popleft()
```

```
if state == goal:
```

```
    print("Solved!")
```

```
    print("Moves:", len(path))
```

```
    print("Final State:", state)
```

```
    break
```

```
if state in visited:
```

```
    continue
```

```
visited.add(state)
```

```
blank = state.index(0)
```

```
for move in moves[blank]:
```

```
    new = list(state)
```

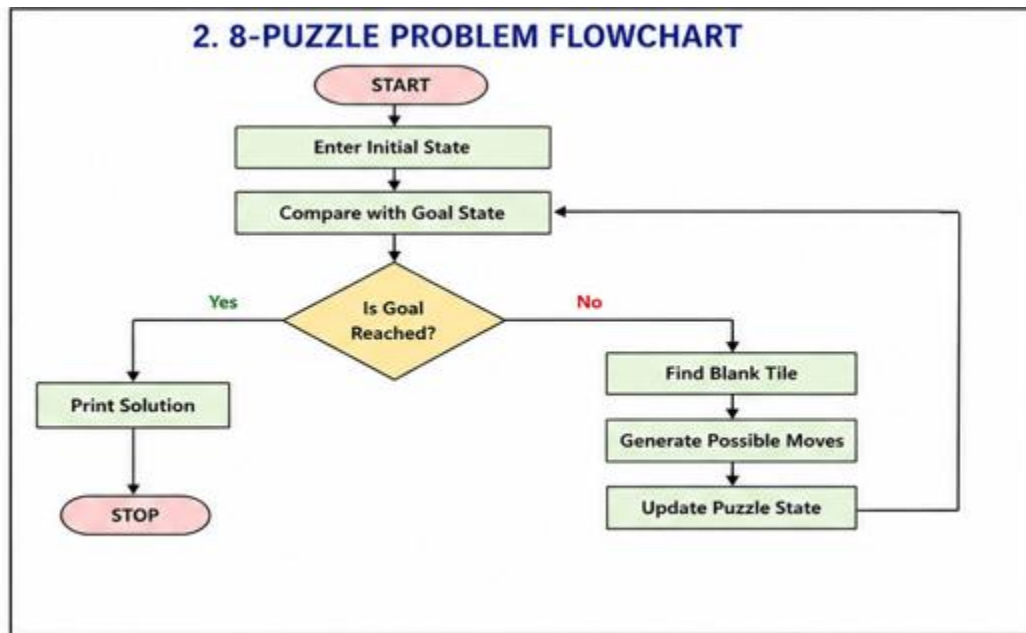
```
    new[blank], new[move] = new[move], new[blank]
```

```
    queue.append((tuple(new), path + [tuple(new)]))
```

```
else:
```

```
    print("No solution found.")
```

Flowchart:



Output:

```
IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Sheik Mohamed/Downloads/water program.py =====
Enter start state (9 numbers separated by space): 1 2 3 4 0 6 7 5 8
Enter goal state (9 numbers separated by space): 1 2 3 4 5 6 7 8 0
Solved!
Moves: 2
Final State: (1, 2, 3, 4, 5, 6, 7, 8, 0)
>>>
```

Result:

Thus, the Python program to solve the 8-Puzzle problem was executed successfully.

